

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 14B

ACID-BASE TITRATIONS, PH CURVES & BUFFER SOLUTIONS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 14: ACID-BASE EQUILIBRIA
SUB-TOPIC:	14B: Acid-Base Titrations, pH Curves & Buffer Solutions
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	229 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

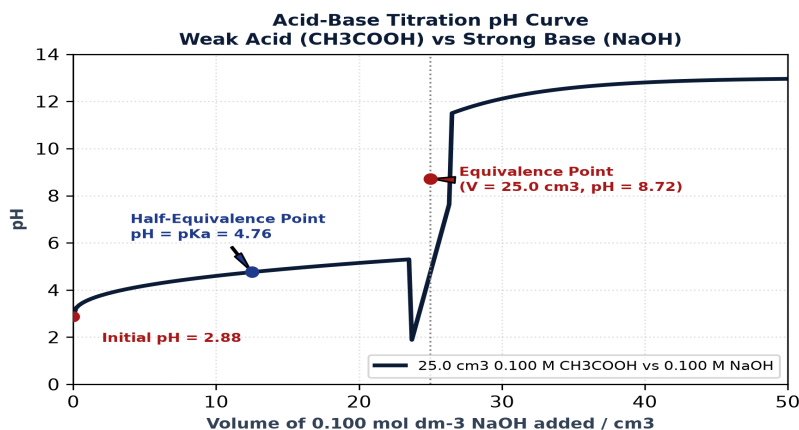
HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25

Q1. 1. Titration pH Curve & Indicator Selection: Weak Acid vs Strong Base**[5 Marks]****WCH14/01/Jan23/Q13**

25.0 cm³ of 0.100 mol dm⁻³ CH₃COOH ($K_a = 1.74 \times 10^{-5}$) is titrated with 0.100 mol dm⁻³ NaOH.



(a) Describe the shape of the pH curve including initial pH, buffer region, half-equivalence pH, and equivalence point pH. [4]

(b) Select a suitable indicator (Phenolphthalein $pK_{in} = 9.3$ vs Methyl Orange $pK_{in} = 3.7$) with justification. [1]

Q2. 2. Buffer Action Mechanism: Acidic Buffer**[4 Marks]****WCH14/01/Oct22/Q14**

An acidic buffer solution contains ethanoic acid, CH₃COOH, and sodium ethanoate, CH₃COONa.

(a) Define a buffer solution. [1]

(b) Explain with ionic equations how this buffer resists pH changes when small amounts of H⁺ or OH⁻ are added. [3]

Q3. 3. Acidic Buffer pH Calculation: Henderson-Hasselbalch**[4 Marks]****WCH14/01/Jun22/Q15**

Calculate the pH of a buffer solution prepared by mixing 500 cm³ of 0.200 M CH₃COOH with 500 cm³ of 0.100 M CH₃COONa at 298 K. ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³, $pK_a = 4.76$)

(a) Calculate moles of CH_3COOH and CH_3COO^- in the mixture. [2]

(b) Calculate the pH using Henderson-Hasselbalch equation $\text{pH} = \text{pK}_a + \log_{10}([\text{A}^-]/[\text{HA}])$. [2]

Q4. 4. Buffer Action after Adding Small Amount of Strong Acid

[5 Marks]

WCH14/01/Jan22/Q16

To 1.00 dm³ of the buffer in Q3 (0.100 mol CH_3COOH and 0.050 mol CH_3COONa), 0.010 mol of solid HCl is added.

(a) Calculate new moles of CH_3COOH and CH_3COO^- . [2]

(b) Calculate the new pH and the pH change. [3]

Q5. 5. Basic Buffer Action & pH Calculation: NH_3 and NH_4Cl

[5 Marks]

WCH14/01/Oct21/Q15

A basic buffer is prepared from 0.200 M NH_3 and 0.100 M NH_4Cl . ($\text{pK}_b = 4.75$, pK_a of $\text{NH}_4^+ = 9.25$)

(a) Write ionic equations for buffer action when H^+ or OH^- is added. [2]

(b) Calculate pH using $\text{pH} = \text{pK}_a + \log_{10}([\text{NH}_3]/[\text{NH}_4^+])$. [3]

Q6. 6. Carbonate Buffer System in Blood Plasma

[4 Marks]

WCH14/01/Jun21/Q16

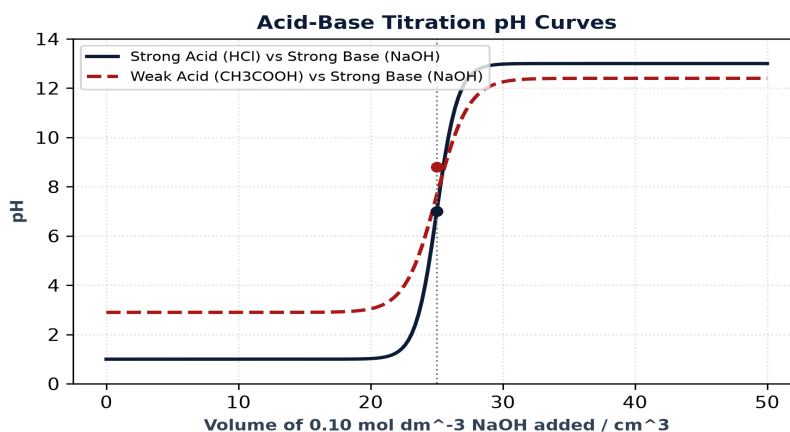
Human blood pH is buffered at 7.40 by the $\text{H}_2\text{CO}_3 / \text{HCO}_3^-$ conjugate pair ($\text{pK}_{a1} = 6.35$).

(a) Calculate the ratio $[\text{HCO}_3^-] / [\text{H}_2\text{CO}_3]$ in blood plasma. [2]

(b) Explain how the kidneys and lungs maintain blood buffer capacity. [2]

Q7. 7. Four Types of Acid-Base Titration pH Curves Comparison**[4 Marks]****WCH14/01/Jan21/Q16**

Compare pH curve features for: (1) Strong Acid + Strong Base, (2) Weak Acid + Strong Base, (3) Strong Acid + Weak Base, (4) Weak Acid + Weak Base.



(a) State equivalence point pH for each of the 4 combinations. [4]

Q8. 8. Half-Equivalence Point Analysis for Weak Acid Titrations**[3 Marks]****WCH14/01/Oct20/Q15**

During titration of weak acid HA with NaOH, half of HA is neutralised at $V = 12.5 \text{ cm}^3$.

(a) Show that at the half-equivalence point, $\text{pH} = \text{pK}_a$. [2]

(b) State how pK_a can be determined directly from a titration curve. [1]

Q9. 9. Preparing a Buffer of Specific Target pH**[4 Marks]****WCH14/01/Jun20/Q16**

A student needs to prepare a buffer of $\text{pH} = 5.00$ using propanoic acid, $\text{C}_2\text{H}_5\text{COOH}$ ($\text{pK}_a = 4.88$).

(a) Calculate the ratio $[\text{C}_2\text{H}_5\text{COO}^-] / [\text{C}_2\text{H}_5\text{COOH}]$ required. [2]

(b) Calculate mass of sodium propanoate $\text{C}_2\text{H}_5\text{COONa}$ ($M_r = 96.0$) to add to 250 cm^3 of 0.200 M acid. [2]

Q10. 10. Buffer Preparation via Partial Neutralisation of Weak Acid with NaOH

[5 Marks]

WCH14/01/Jan20/Q16

100 cm^3 of 0.500 M CH_3COOH is mixed with 50.0 cm^3 of 0.400 M NaOH . ($\text{pK}_a = 4.76$)

(a) Calculate moles of CH_3COOH remaining and CH_3COO^- formed. [3]

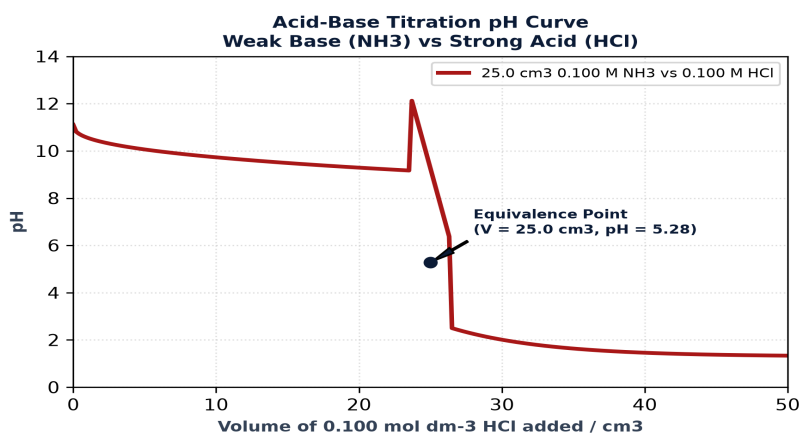
(b) Calculate the pH of the resulting buffer. [2]

Q11. 11. Titration Curve: Weak Base vs Strong Acid

[5 Marks]

WCH14/01/Oct19/Q16

25.0 cm^3 of 0.100 M NH_3 ($K_b = 1.74 \times 10^{-5}$, $\text{pK}_b = 4.75$) is titrated with 0.100 M HCl .



(a) Describe the shape of the pH curve. [4]

(b) Select a suitable indicator (Methyl Red $\text{pK}_{\text{in}} = 5.1$ vs Phenolphthalein $\text{pK}_{\text{in}} = 9.3$). [1]

Q12. 12. Buffer Action after Adding Small Amount of Strong Base**[5 Marks]**
WCH14/01/Jun19/Q17

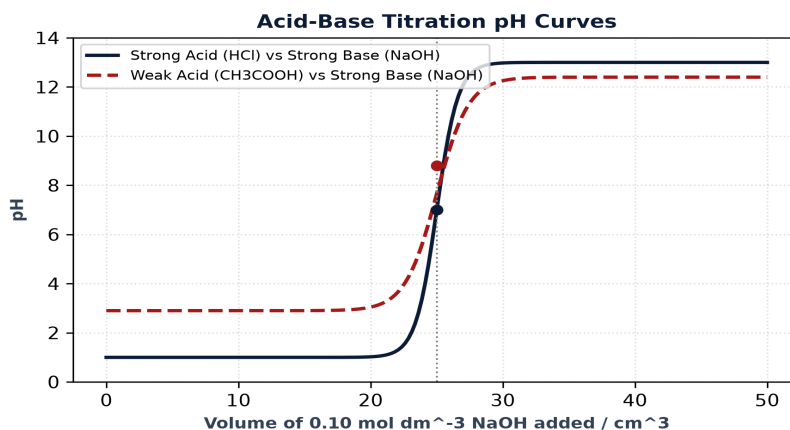
To 500 cm³ of buffer containing 0.200 M CH₃COOH and 0.200 M CH₃COONa (pK_a = 4.76), 0.010 mol NaOH is added.

(a) Calculate new moles of CH₃COOH and CH₃COO⁻. [2]

(b) Calculate new pH. [3]

Q13. 13. Polyprotic Acid Titration Curves: Carbonic Acid H₂CO₃**[4 Marks]**
WCH14/01/Jan19/Q18

H₂CO₃ has two dissociation steps (K_{a1} = 4.5 × 10⁻⁷, K_{a2} = 4.7 × 10⁻¹¹).



(a) Describe the pH curve for titrating H₂CO₃ with NaOH (2 equivalence points). [4]

Q14. 14. Enthalpy Profile of Buffer Actions**[3 Marks]**
WCH14/01/sample/Q14

Explain why adding small amounts of acid or base to a buffer causes almost no change in temperature or pH.

(a) Relate buffer capacity to reservoir concentrations. [3]

Q15. 15. Indicator Transition Range & pK_{in} Formula**[4 Marks]****WCH14/01/Sample/Q15**

An indicator HIn has pK_{in} = 3.70 (Methyl Orange).

- (a) Write the HIn dissociation equation and K_{in} expression. [2]

- (b) Explain why indicator colour change spans a range of ~2 pH units (pK_{in} ± 1). [2]

Q16. 16. Buffer Action in Industrial Fermentation**[4 Marks]****WCH14/01/Sample/Q16**

Enzyme-catalysed fermentation requires strict pH maintenance at 6.80.

- (a) Select a suitable phosphate buffer system H₂PO₄⁻ / HPO₄²⁻ (pK_{a2} = 7.21). [2]

- (b) Calculate the ratio [HPO₄²⁻] / [H₂PO₄⁻] needed for pH 6.80. [2]

Q17. 17. Titration of Diprotic Acid Oxalic Acid (COOH)₂**[4 Marks]****WCH14/01/Sample/Q17**

25.0 cm³ of 0.050 M (COOH)₂ is titrated with 0.100 M NaOH.

- (a) Calculate total volume of NaOH needed for complete neutralisation. [2]

- (b) Explain why only one steep vertical section is observed. [2]

Q18. 18. Buffer Capacity Definition & Factors Affecting It**[4 Marks]****WCH14/01/Sample/Q18**

Buffer A: 1.0 M CH₃COOH / 1.0 M CH₃COONa; Buffer B: 0.01 M CH₃COOH / 0.01 M CH₃COONa.

- (a) Compare the initial pH of Buffer A and Buffer B. [1]

(b) Compare the buffer capacity of Buffer A and Buffer B. [3]

Q19. 19. Selection of Indicators for Weak Acid – Weak Base Titration

[3 Marks]

WCH14/01/sample/Q19

For 25.0 cm³ of 0.100 M CH₃COOH vs 0.100 M NH₃.

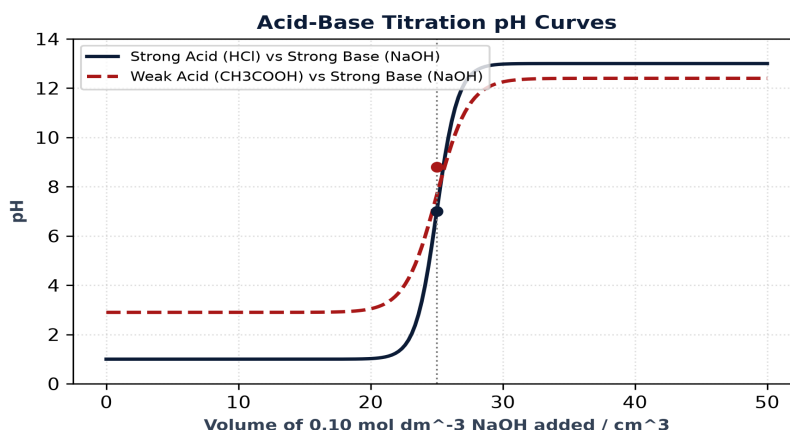
(a) Explain why NO indicator is suitable for a weak acid – weak base titration. [3]

Q20. 20. Titration Curve Construction: Strong Base into Strong Acid

[4 Marks]

WCH14/01/sample/Q20

25.0 cm³ of 0.100 M HCl titrated with 0.100 M NaOH.



(a) Calculate pH at: (i) $V = 0$ cm³, (ii) $V = 24.9$ cm³, (iii) $V = 25.0$ cm³, (iv) $V = 25.1$ cm³. [4]

Q21. 21. Buffer pH Calculation with Partial Neutralisation of Ammonia

[5 Marks]

WCH14/01/sample/Q21

100 cm³ of 0.200 M NH₃ is mixed with 50.0 cm³ of 0.100 M HCl. (pK_a of NH₄⁺ = 9.25)

(a) Calculate moles of NH₃ remaining and NH₄⁺ formed. [3]

(b) Calculate final buffer pH. [2]

Q22. 22. Blood Buffer Action during Strenuous Exercise (Lactic Acid)

[4 Marks]

WCH14/01/Sample/Q22

Strenuous exercise releases lactic acid (HLac) into blood plasma.

(a) Write an ionic equation showing how HCO_3^- buffers lactic acid. [2]

(b) Explain how the respiratory system responds to restore blood pH. [2]

Q23. 23. Calculations of Salt/Acid Ratio from Target Buffer pH

[4 Marks]

WCH14/01/Sample/Q23

Target buffer pH = 4.50 using ethanoic acid ($\text{pK}_a = 4.76$).

(a) Calculate the ratio $[\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$. [2]

(b) Calculate volume of 0.200 M CH_3COONa to add to 100 cm³ of 0.100 M CH_3COOH . [2]

Q24. 24. Phenolphthalein Endpoint Mechanism

[3 Marks]

WCH14/01/Sample/Q24

Phenolphthalein is colourless in acid (HIn) and pink in alkali (In⁻).

(a) Explain using Le Chatelier's principle why phenolphthalein turns pink when excess NaOH is added during titration. [3]

Q25. 25. Methyl Orange Endpoint Mechanism

[3 Marks]

WCH14/01/Sample/Q25

Methyl orange is red in acid (HIn), yellow in alkali (In^-), and orange at endpoint.

(a) State the pH at which methyl orange appears orange ($[\text{HIn}] = [\text{In}^-]$). [1]

(b) Why is methyl orange suitable for titrating Na_2CO_3 with HCl ? [2]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Double End-Point Titration of Na_2CO_3 and NaHCO_3 Mixture****[6 Marks]**
WCH14/01/Hard/Q26

25.0 cm³ mixture of Na_2CO_3 and NaHCO_3 titrated with 0.100 M HCl.

Phenolphthalein endpoint reached at 12.0 cm³ HCl. Methyl orange endpoint reached at additional 18.0 cm³ HCl.

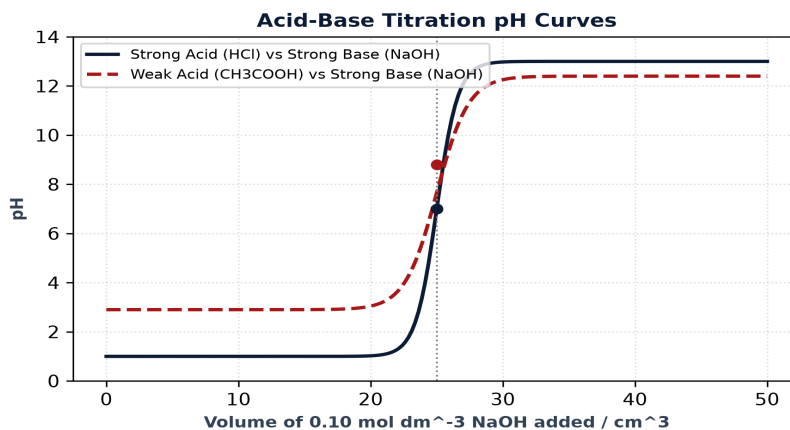
(a) Write equations for reactions occurring at each endpoint. [2]

(b) Calculate concentrations of Na_2CO_3 and NaHCO_3 in original mixture. [4]

Q27. 27. A* Challenge: Complete pH Curve Derivation for Weak Acid – Strong Base Titration**[6 Marks]**
WCH14/01/Hard/Q27

25.0 cm³ of 0.200 M CH_3COOH ($\text{pK}_a = 4.76$) titrated with 0.200 M NaOH.

Calculate pH at: $V = 0$, $V = 12.5$, $V = 25.0$, $V = 30.0$ cm³.



(a) Calculate initial pH ($V = 0$ cm³). [1]

(b) Calculate half-equivalence pH ($V = 12.5$ cm³). [1]

(c) Calculate equivalence point pH ($V = 25.0$ cm³). [2]

(d) Calculate post-equivalence pH ($V = 30.0$ cm³). [2]

Q28. 28. A* Challenge: Buffer Resistivity to Dilution**[5 Marks]**
WCH14/01/Hard/Q28

A buffer contains 0.100 M CH_3COOH and 0.100 M CH_3COONa ($\text{pH} = 4.76$).
The buffer is diluted 10-fold with pure water.

(a) Calculate the new pH after 10-fold dilution. [3]

(b) Explain why diluting a buffer does NOT alter its pH. [2]

Q29. 29. A* Challenge: Buffer Capacity Derivation and Maximum Buffer Region**[5 Marks]**
WCH14/01/Hard/Q29

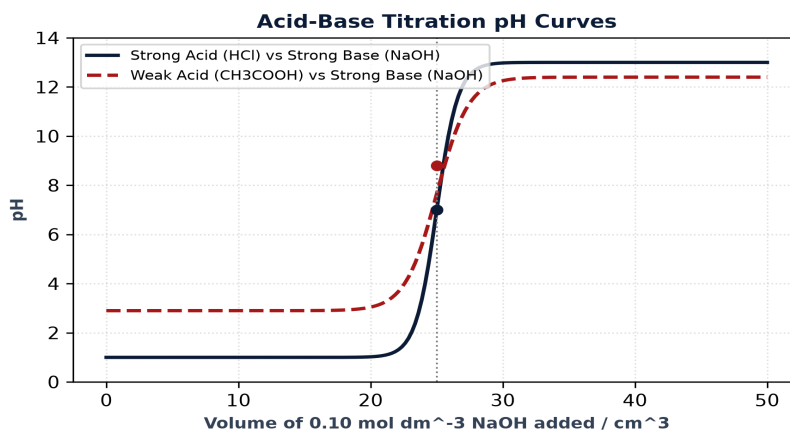
Buffer capacity $\beta = \text{dB} / \text{dpH}$ is maximized when $[\text{A}^-] = [\text{HA}]$.

(a) Prove that buffer capacity is highest when $\text{pH} = \text{pK}_a$. [3]

(b) State the effective buffering pH range for any weak acid conjugate pair. [2]

Q30. 30. A* Challenge: Amino Acid Titration Curve & Zwitterion Isoelectric Point**[5 Marks]**
WCH14/01/Hard/Q30

Titration of 25.0 cm³ 0.100 M Glycine hydrochloride ($+\text{H}_3\text{N}-\text{CH}_2-\text{COOH}$) with 0.100 M NaOH.
 $\text{pK}_{a1} = 2.34$ ($-\text{COOH}$), $\text{pK}_{a2} = 9.60$ ($-\text{NH}_3^+$).



(a) Sketch the two-stage pH titration curve showing two equivalence points. [3]

(b) Calculate the isoelectric point $pI = (2.34 + 9.60)/2 = 5.97$. [2]

Q31. 31. A* Challenge: Multi-Component Buffer Solution pH Calculation

[5 Marks]
WCH14/01/Hard/Q31

A buffer solution contains 0.10 M HCOOH ($pK_a = 3.75$), 0.10 M CH_3COOH ($pK_a = 4.76$), and 0.10 M HCOONa .

(a) Calculate the pH of the mixture considering HCOOH as primary proton donor. [3]

(b) Calculate concentration of un-ionized CH_3COOH . [2]

Q32. 32. A* Challenge: Buffer Action in Ocean Carbonate Chemistry

[5 Marks]
WCH14/01/Hard/Q32

Ocean water is buffered at pH ~8.1 by $\text{HCO}_3^- / \text{CO}_3^{2-}$ system ($pK_a = 10.33$).

(a) Calculate ratio $[\text{CO}_3^{2-}] / [\text{HCO}_3^-]$ at ocean pH 8.1. [3]

- (b) Explain why falling ocean pH reduces availability of CO_3^{2-} ions for calcifying organisms (mollusks/corals). [2]

Q33. 33. A* Challenge: Polyprotic Acid Titration: Phosphoric Acid H_3PO_4 with NaOH

[5 Marks]
WCH14/01/Hard/Q33

25.0 cm³ of 0.100 M H_3PO_4 titrated with 0.100 M NaOH .
 $K_{a1} = 7.1 \times 10^{-3}$, $K_{a2} = 6.3 \times 10^{-8}$, $K_{a3} = 4.5 \times 10^{-13}$.

- (a) Calculate volumes of NaOH for the 1st, 2nd, and 3rd equivalence points. [3]

- (b) Explain why only the first two equivalence points can be observed with indicators in water. [2]

Q34. 34. A* Challenge: Exact Indicator Color Change Calculation

[5 Marks]
WCH14/01/Hard/Q34

Bromothymol blue (HIn yellow, In⁻ blue, $pK_{in} = 7.10$) is added to a solution.

- (a) Calculate percentage of indicator in blue form (In⁻) at pH 6.50, pH 7.10, and pH 7.80. [4]

- (b) State the visible colour at each pH. [1]

Q35. 35. A* Challenge: Enthalpy of Neutralisation of Polyprotic Acids

[5 Marks]
WCH14/01/Hard/Q35

Titration of H_3PO_4 with NaOH releases 156.9 kJ per mole of H_3PO_4 .

- (a) Calculate enthalpy of neutralisation per mole of OH^- reacted. [2]

- (b) Compare with strong acid value ($-57.1 \text{ kJ mol}^{-1}$) and explain difference. [3]

Q36. 36. A* Challenge: Titration of Weak Base Ammonia with Strong Acid HNO₃**[5 Marks]**
WCH14/01/Hard/Q36

50.0 cm³ of 0.150 M NH₃ (pK_b = 4.75) titrated with 0.150 M HNO₃.

(a) Calculate initial pH. [1]

(b) Calculate pH at half-equivalence (V = 25.0 cm³). [1]

(c) Calculate pH at equivalence point (V = 50.0 cm³). [3]

Q37. 37. A* Challenge: Buffer Capacity Breakdown at Extreme pH**[5 Marks]**
WCH14/01/Hard/Q37

To 100 cm³ of 0.100 M CH₃COOH / 0.100 M CH₃COONa buffer (pH = 4.76), 0.120 mol of NaOH is added.

(a) Explain why the buffer capacity is exceeded. [2]

(b) Calculate the final pH. [3]

Q38. 38. A* Challenge: Universal Indicator Spectral RGB pH Mapping**[4 Marks]**
WCH14/01/Hard/Q38

Universal indicator is a mixture of water-soluble indicators (thymol blue, methyl red, bromothymol blue, phenolphthalein).

(a) Explain why universal indicator exhibits a continuous spectrum of colours across pH 1–14. [4]

Q39. 39. A* Challenge: pH Calculation of Mixed Weak Acid and Strong Base Buffer System**[5 Marks]**
WCH14/01/Hard/Q39

200 cm³ of 0.300 M HCOOH (pK_a = 3.75) is mixed with 100 cm³ of 0.400 M KOH.

(a) Calculate moles of HCOOH remaining and HCOO⁻ formed. [2]

(b) Calculate buffer pH. [3]

Q40. 40. A* Challenge: Determination of Equivalence Point via Conductometric Titration**[5 Marks]**
WCH14/01/Hard/Q40

Titration of 25.0 cm³ HCl with NaOH monitored by electrical conductivity.

(a) Explain why conductivity decreases to a minimum at equivalence point and then increases. [3]

(b) Compare with conductivity curve for titrating CH₃COOH with NaOH. [2]

Q41. 41. A* Challenge: Temperature Jump Effects on Buffer pH**[5 Marks]**
WCH14/01/Hard/Q41

A phosphate buffer (pK_a = 7.21 at 298 K, ΔH_{diss} = +4.0 kJ mol⁻¹) is heated to 310 K.

(a) Calculate pK_a at 310 K using van 't Hoff equation. [3]

(b) Calculate pH shift for a 1:1 buffer. [2]

Q42. 42. A* Challenge: Quantitative Buffer Capacity Formula Derivation
[5 Marks]
WCH14/01/Hard/Q42

Buffer capacity $\beta = 2.303 \times c \times \left(\frac{K_a [H^+]}{(K_a + [H^+])^2} \right)$ where $c = [HA] + [A^-]$.

(a) Show that β is maximized when $[H^+] = K_a$. [3]

(b) Calculate maximum buffer capacity $\beta_{\max}$ for a 0.500 M buffer. [2]

Q43. 43. A* Challenge: Titration of Sodium Carbonate with Hydrochloric Acid (2 Steps)
[5 Marks]
WCH14/01/Hard/Q43

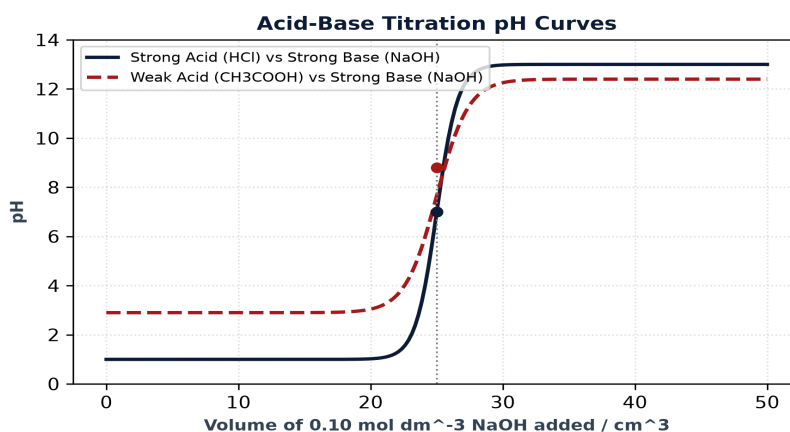
25.0 cm³ of 0.100 M Na₂CO₃ titrated with 0.100 M HCl.

(a) Calculate volumes of HCl for 1st equivalence point (Phenolphthalein) and 2nd (Methyl Orange). [3]

(b) Write balanced equations for both titration steps. [2]

Q44. 44. A* Challenge: Thermometric Titration Curve Analysis
[5 Marks]
WCH14/01/Hard/Q44

25.0 cm³ of 1.00 M HCl in a vacuum flask titrated with 1.00 M NaOH; temperature recorded every 1.0 cm³.



(a) Describe the shape of the temperature vs volume of NaOH curve. [3]

(b) Explain how the equivalence point is identified from the temperature peak. [2]

Q45. 45. A* Challenge: pH of Buffer after Adding Gaseous Ammonia

[5 Marks]
WCH14/01/Hard/Q45

500 cm³ buffer containing 0.200 M HCOOH and 0.200 M HCOONa ($pK_a = 3.75$).
0.020 mol of gaseous NH₃ is dissolved in the buffer.

(a) Write ionic equation for reaction between NH₃ and buffer component. [2]

(b) Calculate final pH. [3]

Q46. 46. A* Challenge: Acidity of Hydrated Copper(II) vs Zinc(II) Ions

[5 Marks]
WCH14/01/Hard/Q46

[Cu(H₂O)₆]²⁺ has $pK_a = 7.3$, while [Zn(H₂O)₆]²⁺ has $pK_a = 9.0$.

(a) Explain why [Cu(H₂O)₆]²⁺ is significantly more acidic than [Zn(H₂O)₆]²⁺ despite both having +2 charge. [3]

(b) Relate this to Jahn–Teller axial distortion in d⁹ Cu²⁺. [2]

Q47. 47. A* Challenge: Buffer Capacity Against Extreme Base Titration

[5 Marks]
WCH14/01/Hard/Q47

250 cm³ buffer containing 0.400 M CH₃COOH and 0.400 M CH₃COONa ($pK_a = 4.76$).

(a) Calculate maximum moles of NaOH that can be added before buffer pH exceeds 5.76 ($pK_a + 1.00$). [5]

Q48. 48. A* Challenge: Titration of Weak Triprotic Acid Citric Acid C₆H₈O₇**[5 Marks]**
WCH14/01/Hard/Q48

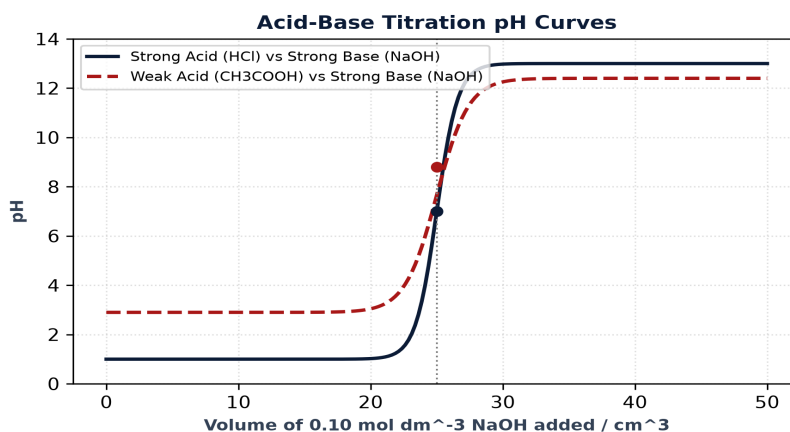
25.0 cm³ of 0.100 M citric acid (triprotic) titrated with 0.100 M NaOH.

(a) Calculate total volume of NaOH to reach final equivalence point. [2]

(b) Explain why citric acid shows a single smooth pH jump at 75.0 cm³. [3]

Q49. 49. A* Challenge: Determination of K_a from pH Curve Derivative dpH/dV**[5 Marks]**
WCH14/01/Hard/Q49

In a computerised titration, dpH/dV is plotted against volume of NaOH added.



(a) Explain why the peak of dpH/dV corresponds to the exact equivalence point. [2]

(b) Explain why the minimum of dpH/dV corresponds to the half-equivalence point. [3]

**Q50. 50. A* Challenge: Complete Titration & Buffer Master
Synthesis****[6 Marks]**
WCH14/01/Hard/Q50

A 50.0 cm³ sample of 0.100 M weak acid HA ($pK_a = 4.00$) is titrated with 0.100 M NaOH.

Calculate pH at: (i) $V = 0$, (ii) $V = 25.0$ cm³, (iii) $V = 50.0$ cm³, (iv) $V = 60.0$ cm³.

(a) Calculate pH at $V = 0$ cm³. [1]

(b) Calculate pH at $V = 25.0$ cm³ (half-equivalence). [1]

(c) Calculate pH at $V = 50.0$ cm³ (equivalence point). [2]

(d) Calculate pH at $V = 60.0$ cm³ (post-equivalence). [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 14B

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Titration pH Curve & Indicator Selection: Weak Acid vs Strong Base 1. (a) Initial pH = 2.88 (1); buffer region rises slowly between 5–20 cm ³ (1); half-equivalence pH = pK _a = 4.76 at 12.5 cm ³ (1); steep vertical jump from pH 7 to 11 with equivalence point at pH = 8.72 at 25.0 cm ³ (1). 1. (b) Phenolphthalein (pK _{in} = 9.3) because its colour change range (8.3–10.0) falls entirely within the vertical pH jump at equivalence point (1).	[5]
Q2	Q2: 2. Buffer Action Mechanism: Acidic Buffer 2. (a) Solution that minimizes pH changes when small amounts of acid or base are added (1). 2. (b) Added H ⁺ reacts with CH ₃ COO ⁻ reservoir: CH ₃ COO ⁻ (aq) + H ⁺ (aq) → CH ₃ COOH(aq) (1.5). Added OH ⁻ reacts with CH ₃ COOH reservoir: CH ₃ COOH(aq) + OH ⁻ (aq) → CH ₃ COO ⁻ (aq) + H ₂ O(l) (1.5).	[4]
Q3	Q3: 3. Acidic Buffer pH Calculation: Henderson-Hasselbalch 3. (a) Moles CH ₃ COOH = 0.500 × 0.200 = 0.100 mol (1); Moles CH ₃ COO ⁻ = 0.500 × 0.100 = 0.050 mol (1). 3. (b) pH = 4.76 + log ₁₀ (0.050 / 0.100) = 4.76 + log ₁₀ (0.50) = 4.76 - 0.301 = 4.46 (2).	[4]
Q4	Q4: 4. Buffer Action after Adding Small Amount of Strong Acid 4. (a) Added H ⁺ reacts with CH ₃ COO ⁻ : Moles CH ₃ COO ⁻ = 0.050 - 0.010 = 0.040 mol (1); Moles CH ₃ COOH = 0.100 + 0.010 = 0.110 mol (1). 4. (b) New pH = 4.76 + log ₁₀ (0.040 / 0.110) = 4.76 - 0.439 = 4.32 (2). Change = 4.32 - 4.46 = -0.14 pH units (minor drop) (1).	[5]
Q5	Q5: 5. Basic Buffer Action & pH Calculation: NH ₃ and NH ₄ Cl 5. (a) Added H ⁺ reacts with NH ₃ : NH ₃ (aq) + H ⁺ (aq) → NH ₄ ⁺ (aq) (1). Added OH ⁻ reacts with NH ₄ ⁺ : NH ₄ ⁺ (aq) + OH ⁻ (aq) → NH ₃ (aq) + H ₂ O(l) (1). 5. (b) pH = 9.25 + log ₁₀ (0.200 / 0.100) = 9.25 + log ₁₀ (2.0) = 9.25 + 0.301 = 9.55 (3).	[5]
Q6	Q6: 6. Carbonate Buffer System in Blood Plasma 6. (a) 7.40 = 6.35 + log ₁₀ (ratio) ⇒ log ₁₀ (ratio) = 1.05 ⇒ Ratio = 10 ^{1.05} = 11.2 (2). 6. (b) Lungs regulate [H ₂ CO ₃] by exhaling CO ₂ gas (1); kidneys regulate [HCO ₃ ⁻] by excreting or reabsorbing bicarbonate ions (1).	[4]
Q7	Q7: 7. Four Types of Acid-Base Titration pH Curves Comparison 7. (a) 1. Strong-Strong: pH = 7.00 (1). 2. Weak Acid-Strong Base: pH > 7 (~8.5–9.0) (1). 3. Strong Acid-Weak Base: pH < 7 (~5.0–5.5) (1). 4. Weak Acid-Weak Base: pH ~ 7 (no steep vertical section) (1).	[4]
Q8	Q8: 8. Half-Equivalence Point Analysis for Weak Acid Titrations 8. (a) At half-equivalence, [A ⁻] = [HA] ⇒ [A ⁻]/[HA] = 1 ⇒ log ₁₀ (1) = 0 ⇒ pH = pK _a + 0 = pK _a (2). 8. (b) Read pH value on vertical axis corresponding to half the equivalence volume on horizontal axis (1).	[3]
Q9	Q9: 9. Preparing a Buffer of Specific Target pH 9. (a) 5.00 = 4.88 + log ₁₀ (ratio) ⇒ log ₁₀ (ratio) = 0.12 ⇒ Ratio = 10 ^{0.12} = 1.318 (2). 9. (b) Moles acid = 0.250 × 0.200 = 0.050 mol. Moles salt required = 1.318 × 0.050 = 0.0659 mol. Mass = 0.0659 × 96.0 = 6.33 g (2).	[4]
Q10	Q10: 10. Buffer Preparation via Partial Neutralisation of Weak Acid with NaOH 10. (a) Initial CH ₃ COOH = 0.050 mol; Added OH ⁻ = 0.020 mol. Reaction: CH ₃ COOH + OH ⁻ → CH ₃ COO ⁻ + H ₂ O. Remaining CH ₃ COOH = 0.030 mol; Formed CH ₃ COO ⁻ = 0.020 mol (3). 10. (b) pH = 4.76 + log ₁₀ (0.020 / 0.030) = 4.76 - 0.176 = 4.58 (2).	[5]
Q11	Q11: 11. Titration Curve: Weak Base vs Strong Acid 11. (a) Initial pH = 11.12 (1); buffer region around pH 9.25 (1); steep vertical jump from pH 7 to 3 at 25.0 cm ³ (1); equivalence point pH = 5.28 (1). 11. (b) Methyl Red (pK _{in} = 5.1) because its colour change range matches acidic equivalence point (1).	[5]

Q12	Q12: 12. Buffer Action after Adding Small Amount of Strong Base 12. (a) Initial moles = 0.100 mol each. Added OH ⁻ reacts with CH ₃ COOH: CH ₃ COOH = 0.090 mol; CH ₃ COO ⁻ = 0.110 mol (2). 12. (b) New pH = 4.76 + log ₁₀ (0.110 / 0.090) = 4.76 + 0.087 = 4.85 (minor rise from 4.76) (3).	[5]
Q13	Q13: 13. Polyprotic Acid Titration Curves: Carbonic Acid H ₂ CO ₃ 13. (a) First step to HCO ₃ ⁻ gives first equivalence point at V ₁ (~ pH 8.3) (2); second step to CO ₃ ²⁻ gives second equivalence point at 2V ₁ (~ pH 11.5) (2).	[4]
Q14	Q14: 14. Enthalpy Profile of Buffer Actions 14. (a) Buffer contains high concentrations of both weak acid HA and conjugate base A ⁻ (1). Added H ⁺ or OH ⁻ is fully neutralized by the large reservoirs (1). Ratio [A ⁻]/[HA] changes negligibly, keeping pH constant (1).	[3]
Q15	Q15: 15. Indicator Transition Range & pK _{in} Formula 15. (a) HIn(aq) ⇌ H ⁺ (aq) + In ⁻ (aq); K _{in} = [H ⁺][In ⁻] / [HIn] (2). 15. (b) Human eye detects colour change when ratio [In ⁻]/[HIn] changes from 1:10 (pH = pK _{in} - 1) to 10:1 (pH = pK _{in} + 1) (2).	[4]
Q16	Q16: 16. Buffer Action in Industrial Fermentation 16. (a) H ₂ PO ₄ ⁻ / HPO ₄ ²⁻ system has pK _{a2} = 7.21, close to target pH 6.80 (2). 16. (b) 6.80 = 7.21 + log ₁₀ (ratio) ⇒ log ₁₀ (ratio) = -0.41 ⇒ Ratio = 10 ^{-0.41} = 0.389 (2).	[4]
Q17	Q17: 17. Titration of Diprotic Acid Oxalic Acid (COOH) ₂ 17. (a) Moles acid = 0.025 x 0.050 = 0.00125 mol ⇒ Moles NaOH = 2 x 0.00125 = 0.00250 mol. Volume = 0.00250 / 0.100 = 0.0250 dm ³ = 25.0 cm ³ (2). 17. (b) K _{a1} and K _{a2} values are relatively close, so two equivalence steps merge into a single steep region (2).	[4]
Q18	Q18: 18. Buffer Capacity Definition & Factors Affecting It 18. (a) Both buffers have identical pH = 4.76 because ratio [A ⁻]/[HA] = 1.0 in both (1). 18. (b) Buffer A has 100x higher buffer capacity (1). It contains 100x larger mole reservoirs of HA and A ⁻ , resisting far larger additions of acid/base before breakdown (2).	[4]
Q19	Q19: 19. Selection of Indicators for Weak Acid - Weak Base Titration 19. (a) There is no steep vertical section on the pH curve near equivalence point (1). The pH changes gradually over several cm ³ of added base (1). No indicator can change colour sharply within 1 drop of added titrant (1).	[3]
Q20	Q20: 20. Titration Curve Construction: Strong Base into Strong Acid 20. (a) (i) V=0: [H ⁺]=0.100 ⇒ pH = 1.00 (1). (ii) V=24.9: excess H ⁺ = 0.0001 mol in 49.9 cm ³ ⇒ [H ⁺]=0.0020 M ⇒ pH = 2.70 (1). (iii) V=25.0: neutral ⇒ pH = 7.00 (1). (iv) V=25.1: excess OH ⁻ = 0.0001 mol in 50.1 cm ³ ⇒ [OH ⁻]=0.0020 M ⇒ pH = 11.30 (1). (Demonstrates steep jump 2.7 → 11.3 across 0.2 cm ³ !).	[4]
Q21	Q21: 21. Buffer pH Calculation with Partial Neutralisation of Ammonia 21. (a) Initial NH ₃ = 0.020 mol; Added H ⁺ = 0.0050 mol. Reaction: NH ₃ + H ⁺ → NH ₄ ⁺ . Remaining NH ₃ = 0.015 mol; Formed NH ₄ ⁺ = 0.0050 mol (3). 21. (b) pH = 9.25 + log ₁₀ (0.015 / 0.0050) = 9.25 + log ₁₀ (3.0) = 9.25 + 0.477 = 9.73 (2).	[5]
Q22	Q22: 22. Blood Buffer Action during Strenuous Exercise (Lactic Acid) 22. (a) HLac(aq) + HCO ₃ ⁻ (aq) → Lac ⁻ (aq) + H ₂ O(l) + CO ₂ (g) (2). 22. (b) Increased CO ₂ and H ⁺ trigger faster breathing rate (hyperpnea), expelling CO ₂ gas and shifting equilibrium H ₂ O + CO ₂ ⇌ H ₂ CO ₃ ⇌ H ⁺ + HCO ₃ ⁻ left to remove H ⁺ (2).	[4]
Q23	Q23: 23. Calculations of Salt/Acid Ratio from Target Buffer pH 23. (a) 4.50 = 4.76 + log ₁₀ (ratio) ⇒ log ₁₀ (ratio) = -0.26 ⇒ Ratio = 0.550 (2). 23. (b) Moles acid = 0.100 x 0.100 = 0.010 mol. Moles salt needed = 0.550 x 0.010 = 0.0055 mol. Volume salt = 0.0055 / 0.200 = 0.0275 dm ³ = 27.5 cm ³ (2).	[4]
Q24	Q24: 24. Phenolphthalein Endpoint Mechanism 24. (a) HIn(colourless) ⇌ H ⁺ + In ⁻ (pink). Added OH ⁻ removes H ⁺ ions forming H ₂ O (1). Equilibrium shifts right to replace H ⁺ ions (1), converting colourless HIn into pink In ⁻ ions (1).	[3]
Q25	Q25: 25. Methyl Orange Endpoint Mechanism 25. (a) At pK _{in} = 3.70 (1). 25. (b) Carbonate titration forms carbonic acid H ₂ CO ₃ (pH ~4 at equivalence point), matching methyl orange transition range (3.1-4.4) (2).	[3]

Q26	Q26: 26. A* Challenge: Double End-Point Titration of Na₂CO₃ and NaHCO₃ Mixture 26. (a) Phenolphthalein: CO₃²⁻ + H⁺ → HCO₃⁻ (1). Methyl orange: HCO₃⁻ + H⁺ → H₂O + CO₂ (1). 26. (b) Phenolphthalein volume V₁ = 12.0 cm³ reacts with ALL Na₂CO₃ ⇒ Moles Na₂CO₃ = 0.0120 × 0.100 = 0.00120 mol ⇒ [Na₂CO₃] = 0.00120 / 0.025 = 0.0480 M (2). Second step V₂ = 18.0 cm³ neutralises HCO₃⁻ from Na₂CO₃ (12.0 cm³) + original NaHCO₃ (6.0 cm³) ⇒ Moles NaHCO₃ = 0.0060 × 0.100 = 0.00060 mol ⇒ [NaHCO₃] = 0.00060 / 0.025 = 0.0240 M (2).	[6]
Q27	Q27: 27. A* Challenge: Complete pH Curve Derivation for Weak Acid – Strong Base Titration 27. (a) V=0: [H⁺] = sqrt(1.74×10⁻⁵ × 0.200) = 1.865×10⁻³ ⇒ pH = 2.73 (1). 27. (b) V=12.5: pH = pK_a = 4.76 (1). 27. (c) V=25.0: [CH₃COO⁻] = 0.0050 / 0.050 = 0.100 M. K_b = 5.75×10⁻¹⁰ ⇒ [OH⁻] = sqrt(5.75×10⁻¹¹) = 7.58×10⁻⁶ M ⇒ pOH = 5.12 ⇒ pH = 8.88 (2). 27. (d) V=30.0: Excess OH⁻ = 0.0050 × 0.200 = 0.0010 mol in 55.0 cm³ ⇒ [OH⁻] = 0.01818 M ⇒ pOH = 1.74 ⇒ pH = 12.26 (2).	[6]
Q28	Q28: 28. A* Challenge: Buffer Resistivity to Dilution 28. (a) New concentrations: [CH₃COOH] = 0.0100 M, [CH₃COO⁻] = 0.0100 M. pH = 4.76 + log₁₀(0.0100 / 0.0100) = 4.76 + log₁₀(1.0) = 4.76 (unchanged!) (3). 28. (b) Both [HA] and [A⁻] are diluted by the exact same factor (10x), so ratio [A⁻]/[HA] remains identical (2).	[5]
Q29	Q29: 29. A* Challenge: Buffer Capacity Derivation and Maximum Buffer Region 29. (a) At [A⁻] = [HA], equal reservoirs of weak acid and conjugate base are available to neutralize both H⁺ and OH⁻ additions equally (2). Fractional change in [A⁻]/[HA] ratio per mole of added acid/base is minimized (1). 29. (b) Effective buffering range = pK_a ± 1.00 pH unit (2).	[5]
Q30	Q30: 30. A* Challenge: Amino Acid Titration Curve & Zwitterion Isoelectric Point 30. (a) Two distinct vertical steps: V₁ = 25 cm³ (neutralises -COOH to zwitterion) and V₂ = 50 cm³ (neutralises -NH₃⁺ to -NH₂) (3). 30. (b) pI = (2.34 + 9.60) / 2 = 5.97 (2).	[5]
Q31	Q31: 31. A* Challenge: Multi-Component Buffer Solution pH Calculation 31. (a) HCOOH is stronger acid (pK_a 3.75 vs 4.76), dominating [H⁺]: pH = pK_a(HCOOH) + log₁₀([HCOO⁻]/[HCOOH]) = 3.75 + log₁₀(0.10/0.10) = 3.75 (3). 31. (b) [H⁺] = 10^{-3.75} = 1.78×10⁻⁴ M. CH₃COOH is suppressed by H⁺ from HCOOH (2).	[5]
Q32	Q32: 32. A* Challenge: Buffer Action in Ocean Carbonate Chemistry 32. (a) 8.1 = 10.33 + log₁₀(ratio) ⇒ log₁₀(ratio) = -2.23 ⇒ Ratio = 10^{-2.23} = 0.00589 (0.59%) (3). 32. (b) Lower pH means higher [H⁺], which reacts with CO₃²⁻: CO₃²⁻ + H⁺ → HCO₃⁻, depleting free carbonate ions needed for CaCO₃ shell formation (2).	[5]
Q33	Q33: 33. A* Challenge: Polyprotic Acid Titration: Phosphoric Acid H₃PO₄ with NaOH 33. (a) V₁ = 25.0 cm³ (H₂PO₄⁻), V₂ = 50.0 cm³ (HPO₄²⁻), V₃ = 75.0 cm³ (PO₄³⁻) (3). 33. (b) K_{a3} is extremely small (4.5×10⁻¹³), so HPO₄²⁻ is too weak an acid in water to give a steep pH jump at V₃ (2).	[5]
Q34	Q34: 34. A* Challenge: Exact Indicator Color Change Calculation 34. (a) At pH 6.50: log([In⁻]/[HIn]) = 6.50 - 7.10 = -0.60 ⇒ ratio = 0.251 ⇒ %In⁻ = 20.1% (1.3). At pH 7.10: ratio = 1.0 ⇒ %In⁻ = 50.0% (1.3). At pH 7.80: log(ratio) = +0.70 ⇒ ratio = 5.01 ⇒ %In⁻ = 83.4% (1.4). 34. (b) pH 6.50: Yellow-green; pH 7.10: Green (endpoint); pH 7.80: Blue (1).	[5]
Q35	Q35: 35. A* Challenge: Enthalpy of Neutralisation of Polyprotic Acids 35. (a) 3 moles of OH⁻ react per mole H₃PO₄: Δ_{neut} H = -156.9 / 3 = -52.3 kJ mol⁻¹ per mole OH⁻ (2). 35. (b) Less exothermic than strong acid (-57.1) because H₃PO₄ is weak; energy is absorbed to break P-O-H bonds for 2nd and 3rd dissociation steps (3).	[5]

Q36	Q36: 36. A* Challenge: Titration of Weak Base Ammonia with Strong Acid HNO ₃ 36. (a) $\text{pOH} = 1/2(4.75 - \log_{10}(0.150)) = 2.79 \Rightarrow \text{pH} = 11.21$ (1). 36. (b) Half-equivalence: $\text{pH} = \text{pK}_a(\text{NH}_4^+) = 14.00 - 4.75 = 9.25$ (1). 36. (c) Equivalence: $[\text{NH}_4^+] = 0.0075 / 0.100 = 0.075 \text{ M}$. $\text{K}_a = 5.75 \times 10^{-10}$. $[\text{H}^+] = \sqrt{5.75 \times 10^{-10} \times 0.075} = 6.57 \times 10^{-6} \text{ M} \Rightarrow \text{pH} = 5.18$ (3).	[5]
Q37	Q37: 37. A* Challenge: Buffer Capacity Breakdown at Extreme pH 37. (a) Buffer contains only 0.010 mol CH ₃ COOH. 0.120 mol OH ⁻ completely neutralises all CH ₃ COOH with 0.110 mol excess OH ⁻ (2). 37. (b) Total volume ~ 100 cm ³ $\Rightarrow [\text{OH}^-] = 0.110 / 0.100 = 1.10 \text{ M}$. $\text{pOH} = -\log_{10}(1.10) = -0.04 \Rightarrow \text{pH} = 14.04$ (3).	[5]
Q38	Q38: 38. A* Challenge: Universal Indicator Spectral RGB pH Mapping 38. (a) Universal indicator combines multiple indicators with overlapping pK _{in} values across pH 1-14 (2). As pH changes, different indicator components transition between acidic and basic forms simultaneously, producing smooth colour shifts (red $\rightarrow$ orange $\rightarrow$ yellow $\rightarrow$ green $\rightarrow$ blue $\rightarrow$ purple) (2).	[4]
Q39	Q39: 39. A* Challenge: pH Calculation of Mixed Weak Acid and Strong Base Buffer System 39. (a) Initial HCOOH = 0.060 mol; Added OH ⁻ = 0.040 mol. Remaining HCOOH = 0.020 mol; Formed HCOO ⁻ = 0.040 mol (2). 39. (b) $\text{pH} = 3.75 + \log_{10}(0.040 / 0.020) = 3.75 + \log_{10}(2.0) = 3.75 + 0.301 = 4.05$ (3).	[5]
Q40	Q40: 40. A* Challenge: Determination of Equivalence Point via Conductometric Titration 40. (a) High conductivity H ⁺ ions are replaced by lower conductivity Na ⁺ ions (conductivity drops) (2). At equivalence point, only Na ⁺ and Cl ⁻ present (minimum). Past equivalence, highly conductive OH ⁻ ions accumulate (conductivity rises) (1). 40. (b) CH ₃ COOH has low initial conductivity; adding NaOH forms CH ₃ COONa (conductivity rises gradually); past equivalence, excess OH ⁻ causes rapid conductivity rise (2).	[5]
Q41	Q41: 41. A* Challenge: Temperature Jump Effects on Buffer pH 41. (a) $\ln(\text{K}_{a310}/\text{K}_{a298}) = (-4000/8.31) * (1/310 - 1/298) = -481.35 * (-1.300 \times 10^{-4}) = +0.06257$ (2). $\text{K}_{a310} = \text{K}_{a298} \times e^{+0.06257} = 1.0645 \text{ K}_{a298} \Rightarrow \text{pK}_{a310} = 7.21 - \log_{10}(1.0645) = 7.18$ (1). 41. (b) For 1:1 buffer, $\text{pH} = \text{pK}_a = 7.18$ (minor drop of 0.03 pH units) (2).	[5]
Q42	Q42: 42. A* Challenge: Quantitative Buffer Capacity Formula Derivation 42. (a) When $[\text{H}^+] = \text{K}_a$, denominator $(\text{K}_a + \text{K}_a)^2 = 4 \text{ K}_a^2$. Fraction = $\text{K}_a^2 / (4 \text{ K}_a^2) = 0.25$ (maximum value of $x/(1+x)^2$) (3). 42. (b) $\beta_{\text{max}} = 2.303 \times 0.500 \times 0.25 = 0.288 \text{ mol dm}^{-3} \text{ per pH unit}$ (2).	[5]
Q43	Q43: 43. A* Challenge: Titration of Sodium Carbonate with Hydrochloric Acid (2 Steps) 43. (a) Step 1 ($\text{CO}_3^{2-} \rightarrow \text{HCO}_3^-$): $V_1 = 25.0 \text{ cm}^3 \text{ HCl (1.5)}$. Step 2 ($\text{HCO}_3^- \rightarrow \text{H}_2\text{O} + \text{CO}_2$): Total $V_2 = 50.0 \text{ cm}^3 \text{ HCl (1.5)}$. 43. (b) Step 1: $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaHCO}_3 + \text{NaCl}$ (1). Step 2: $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ (1).	[5]
Q44	Q44: 44. A* Challenge: Thermometric Titration Curve Analysis 44. (a) Temperature rises linearly as exothermic neutralisation $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ proceeds (2). Reaches a sharp maximum peak at equivalence point (25.0 cm ³) (1). Past equivalence, temperature drops gradually as cooler NaOH titrant is added (1). 44. (b) Intersection of the two linear temperature slopes gives exact equivalence volume (2).	[5]
Q45	Q45: 45. A* Challenge: pH of Buffer after Adding Gaseous Ammonia 45. (a) $\text{NH}_3(\text{g}) + \text{HCOOH}(\text{aq}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{HCOO}^-(\text{aq})$ (2). 45. (b) Initial moles = 0.100 mol each. NH ₃ neutralises 0.020 mol HCOOH: HCOOH = 0.080 mol; HCOO ⁻ = 0.120 mol. $\text{pH} = 3.75 + \log_{10}(0.120 / 0.080) = 3.75 + 0.176 = 3.93$ (3).	[5]
Q46	Q46: 46. A* Challenge: Acidity of Hydrated Copper(II) vs Zinc(II) Ions 46. (a) Cu ²⁺ (d ⁹) undergoes Jahn-Teller distortion, shortening two equatorial Cu-O bonds and increasing local charge density on equatorial water ligands (3). 46. (b) Shorter Cu-O bonds polarize equatorial O-H bonds more strongly, making $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ a stronger proton donor (2).	[5]

Q47	Q47: 47. A* Challenge: Buffer Capacity Against Extreme Base Titration 47. (a) Initial moles = 0.100 mol each. Target pH = 5.76 $\Rightarrow$ $5.76 = 4.76 + \log_{10}([A^-]/[HA]) \Rightarrow \log_{10}(\text{ratio}) = 1.00 \Rightarrow \text{ratio} = 10.0$ (2). Let x = moles OH ⁻ added. Moles A ⁻ = 0.100 + x; Moles HA = 0.100 - x (1). $(0.100 + x)/(0.100 - x) = 10.0 \Rightarrow 0.100 + x = 1.00 - 10x \Rightarrow 11x = 0.900 \Rightarrow x = 0.0818$ mol NaOH (2).	[5]
Q48	Q48: 48. A* Challenge: Titration of Weak Triprotic Acid Citric Acid C ₆ H ₈ O ₇ 48. (a) Moles acid = 0.0025 mol $\Rightarrow$ Moles NaOH = 3 x 0.0025 = 0.0075 mol. Volume = 0.0075 / 0.100 = 0.0750 dm ³ = 75.0 cm ³ (2). 48. (b) pK _a values (3.13, 4.76, 6.40) are close together (<2 units apart), so individual buffer regions and equivalence steps overlap into one combined curve (3).	[5]
Q49	Q49: 49. A* Challenge: Determination of K _a from pH Curve Derivative dpH/dV 49. (a) dpH/dV is the rate of pH change. The pH curve is steepest (inflection point) at equivalence, so dpH/dV reaches a maximum peak (2). 49. (b) At half-equivalence, buffer capacity is maximized, so pH changes most slowly per drop of added base (dpH/dV is at a minimum) (3).	[5]
Q50	Q50: 50. A* Challenge: Complete Titration & Buffer Master Synthesis 50. (a) V=0: $[H^+] = \sqrt{1.0 \times 10^{-4} \times 0.100} = 3.16 \times 10^{-3} \text{ M} \Rightarrow \text{pH} = 2.50$ (1). 50. (b) V=25: pH = pK _a = 4.00 (1). 50. (c) V=50: $[A^-] = 0.0050 / 0.100 = 0.050 \text{ M}$. $K_h = 1.00 \times 10^{-14} / 1.0 \times 10^{-4} = 1.0 \times 10^{-10}$. $[OH^-] = \sqrt{1.0 \times 10^{-10} \times 0.050} = 2.236 \times 10^{-6} \text{ M} \Rightarrow \text{pOH} = 5.65 \Rightarrow \text{pH} = 8.35$ (2). 50. (d) V=60: Excess OH ⁻ = 0.010 x 0.100 = 0.0010 mol in 110 cm ³ $\Rightarrow [OH^-] = 0.00909 \text{ M} \Rightarrow \text{pOH} = 2.04 \Rightarrow \text{pH} = 11.96$ (2).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 14B

FAQ #1: Selecting Indicators for Titrations

Category: Indicator Selection

COMMON EXAMINER TRAP / MISTAKE:

Choosing an indicator based on initial pH instead of equivalence point pH.

MODEL ANSWER / EXAMINER EXPECTATION:

Indicator selection depends ONLY on the vertical pH jump at the EQUIVALENCE POINT. Choose an indicator whose pK_{in} falls within the steep vertical region.

FAQ #2: Half-Equivalence Point Property

Category: Half-Equivalence
Shortcut**COMMON EXAMINER TRAP / MISTAKE:**

Calculating half-equivalence pH using complex K_a quadratic formulas.

MODEL ANSWER / EXAMINER EXPECTATION:

At half-equivalence, exactly half of HA is converted to A^- , so $[A^-] = [HA]$. The Henderson-Hasselbalch equation simplifies to $pH = pK_a + \log_{10}(1) = pK_a$.

FAQ #3: Equivalence Point pH for Weak Acid – Strong Base

Category: Equivalence pH Trap

COMMON EXAMINER TRAP / MISTAKE:

Assuming equivalence point pH is always 7.00 for all titrations.

MODEL ANSWER / EXAMINER EXPECTATION:

For weak acid + strong base, the conjugate base A^- hydrolyses in water ($A^- + H_2O \rightleftharpoons HA + OH^-$), generating excess OH^- ions $\Rightarrow$ equivalence point pH is ALKALINE ($pH > 7$).

FAQ #4: Equivalence Point pH for Strong Acid – Weak Base

Category: Equivalence pH Trap

COMMON EXAMINER TRAP / MISTAKE:

Assuming strong acid + weak base equivalence point pH is 7.00.

MODEL ANSWER / EXAMINER EXPECTATION:

For strong acid + weak base, conjugate acid BH^+ hydrolyses ($BH^+ + H_2O \rightleftharpoons B + H_3O^+$), generating excess H^+ ions $\Rightarrow$ equivalence point pH is ACIDIC ($pH < 7$).

FAQ #5: Henderson-Hasselbalch Ratio Flip

Category: Formula Trap

COMMON EXAMINER TRAP / MISTAKE:

Flipping ratio to $\log_{10}([HA]/[A^-])$ in Henderson-Hasselbalch equation.

MODEL ANSWER / EXAMINER EXPECTATION:

The correct Henderson-Hasselbalch formula is $pH = pK_a + \log_{10}([BASE]/[ACID]) = pK_a + \log_{10}([A^-]/[HA])$. If you use $[HA]/[A^-]$, you MUST subtract the log term.

FAQ #6: Adding Acid/Base to Buffer Reservoir Shift

Category: Buffer Calculation
Trap**COMMON EXAMINER TRAP / MISTAKE:**

Adding moles of H^+ to BOTH $[HA]$ and $[A^-]$ in buffer calculations.

MODEL ANSWER / EXAMINER EXPECTATION:

Added H^+ REACTS WITH $[\text{A}^-]$ (decreases $[\text{A}^-]$, increases $[\text{HA}]$). Added OH^- REACTS WITH $[\text{HA}]$ (decreases $[\text{HA}]$, increases $[\text{A}^-]$).

FAQ #7: Diluting a Buffer Solution**Category: Buffer Properties****COMMON EXAMINER TRAP / MISTAKE:**

Thinking diluting a buffer by 10x changes its pH by 1.00 unit.

MODEL ANSWER / EXAMINER EXPECTATION:

Diluting a buffer changes $[\text{HA}]$ and $[\text{A}^-]$ by the exact same ratio factor. The ratio $[\text{A}^-]/[\text{HA}]$ is UNCHANGED, so buffer pH remains CONSTANT.

FAQ #8: Buffer Capacity Limits**Category: Capacity Overload****COMMON EXAMINER TRAP / MISTAKE:**

Assuming a buffer maintains constant pH regardless of how much acid is added.

MODEL ANSWER / EXAMINER EXPECTATION:

A buffer only works while added H^+ or OH^- is LESS than the moles of A^- or HA present in the reservoir. Once exceeded, buffer capacity breaks down and pH shifts drastically.

FAQ #9: Weak Acid – Weak Base Titration Indicators**Category: Titration Limitations****COMMON EXAMINER TRAP / MISTAKE:**

Searching for an indicator for weak acid + weak base titration.

MODEL ANSWER / EXAMINER EXPECTATION:

NO indicator is suitable for weak acid + weak base titrations because there is NO steep vertical region on the pH curve near equivalence.

FAQ #10: Polyprotic Acid Equivalence Volumes**Category: Polyprotic Stoichiometry****COMMON EXAMINER TRAP / MISTAKE:**

Assuming all equivalence steps require different volumes of titrant.

MODEL ANSWER / EXAMINER EXPECTATION:

For a polyprotic acid H_nA , each successive equivalence step requires the EXACT SAME VOLUME of base ($V_1 = V_2 = V_3$).